

COTTON DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In field pathology, making an accurate diagnosis requires separating biotic (infectious pathogens like fungi, viruses, and bacteria) from abiotic factors (non-infectious environmental issues like salinity, nutrient imbalances, or drought). Agronomists utilize a structured process based on field-level macro patterns down to cellular micro-symptoms before drafting a recommendation.

2. Agronomist Investigative Questionnaire

Category I: Variety & Crop Phenology (The Genetic & Structural Baseline)

- What specific cotton variety did you sow, and is it a Bt or non-Bt variety? (Helps identify genetic resistance or known vulnerabilities to specific regional strains).
- What is the exact age of the crop, or how many Days After Sowing (DAS) is it? (Certain diseases target specific stages, e.g., seedling damping-off vs. mid-season wilts).
- What was the germination percentage, and did you notice any early seedling mortality? (Points toward seed-borne or early soil-borne fungal pathogens).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- When exactly did you first notice these symptoms in the field? (Establishes speed of disease progression; rapid shifts indicate physiological issues, while slow spreading points to biological pathogens).
- Are the symptoms distributed uniformly across the entire field, or are they appearing in isolated patches or low-lying areas? (Uniform damage points to abiotic stresses; localized patches point to soil-borne fungi, wilt pockets, or nematodes).
- Are the affected plants close to the field edges, water channels, or near weed clusters? (Edge-effects often indicate viral diseases like CLCuV moving into the field via insect vectors hosted on surrounding weeds).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- Look closely at the leaves: Are they curling upward or downward, and do you see thick green veins or leaf-like growths (enations) underneath? (Classic diagnostic indicators for Cotton Leaf Curl Virus).
- If there are spots on the leaves, what shape are they? Are they rounded with concentric rings, or are they restricted by the leaf veins, looking angular and water-soaked? (Angular water-soaked spots indicate Bacterial Leaf Blight; circular rings suggest fungal spots like Alternaria).
- Where on the plant did the symptoms start—on the older, lower leaves first, or on the fresh, top leaves? (Fungal wilts and mobile nutrient deficiencies show on lower leaves first; viruses and immobile deficiencies show on new growth).
- If you pull up a wilting plant and slice open the main stem vertically, what color is the internal wood/pith? Is it clean white or stained brown/black? (A brown discolored vascular ring confirms a fungal wilt complex like Fusarium or Verticillium; clean white indicates physiological para-wilt).

Category IV: Water, Soil & Root Architecture

- When was your last irrigation, what method did you use (flood vs. drip), and did water stand in the field for more than 24 hours? (Prolonged waterlogging suffocates roots, inducing root rot or physiological drooping).
- What is the typical pH and EC (salinity) of your soil or tube-well water? (High pH >8.0 can lock up micronutrients like iron or zinc, mimicking viral yellowing).
- Is the soil in the affected patch heavily compacted, or is there a hardpan below the surface? (Compaction restricts taproot elongation, causing severe drought/wilt symptoms even if topsoil is damp).

Category V: Crop Nutrition & Chemical Input History

- How many bags of Nitrogen (Urea), Phosphorus (DAP), and Potassium (SOP/MOP) have you applied so far? (Excessive nitrogen creates a lush, succulent canopy highly vulnerable to sucking pests and blight; potassium deficiency causes leaf margin scorch).
- Have you sprayed any weedicides (herbicides) recently? If yes, what was the chemical, the dose, and did you use a completely washed tank? (Herbicide drift or residue causes leaf curling, cupping, and twisting frequently mistaken for virus damage).

Category VI: Environmental Microclimate & Pest Vectors

- What has the weather been like over the past 7 to 10 days? Has it been hot and dry, or cloudy with intermittent rain and high humidity? (High humidity with splashing rain drives fungal/bacterial spore dispersal; hot, dry spells trigger insect vector surges).
- What is the current population of sucking pests, specifically whiteflies, jassids, or thrips, in your field? (Whiteflies are vectors for CLCuV; heavy jassid feeding causes hopper burn which mimics fungal leaf damage).

Category VII: Historical Cultivation & Previous Actions

- What crop was grown in this exact field during the last winter (Rabi) season? (Certain crops host identical soil fungi or nematodes, compounding disease pressure).
- Did you treat the cotton seeds with a fungicide/insecticide protectant before sowing? (Seed treatments delay or prevent early-season vascular wilts and seedling blights).
- Have you already sprayed any fungicide or bio-stimulant to cure this specific problem, and did the plant show any recovery after it? (Prevents chemical duplication and avoids phytotoxicity from over-spraying).

SUGARCANE DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In sugarcane pathology, making an accurate diagnosis requires separating biotic (infectious pathogens like fungi, viruses, bacteria, and phytoplasmas) from abiotic factors (non-infectious environmental issues like waterlogging, nutrient imbalances, or soil salinity). Because sugarcane is a long-duration, vegetatively propagated crop, agronomists utilize a structured process based on field-level macro patterns down to stalk and cellular micro-symptoms before drafting a recommendation.

2. Agronomist Investigative Questionnaire

Category I: Variety & Crop Phenology (The Genetic & Structural Baseline)

- **What specific sugarcane variety was planted, and is it a Plant Crop or a Ratoon Crop?** (Ratoon crops have exponentially higher disease pressure for systemic issues like Ratoon Stunting Disease (RSD) and Smut. Knowing the variety highlights known susceptibilities, especially to Red Rot).
- **What is the exact age of the crop (Months After Planting/Ratooning)?** (Diseases target specific stages: e.g., Pineapple disease affects early sett germination, Pokkah Boeng hits during rapid monsoon growth, and Red Rot often peaks near maturity).
- **What was the initial sett germination percentage, and did you notice rotting at the ends of the setts?** (Poor germination with black/red discoloration inside the un-germinated setts points strongly to sett-borne fungal pathogens like Pineapple disease or early Red Rot).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- **When exactly did you first notice the yellowing, drying, or stunting in the field?** (Rapid, uniform drying often points to sudden abiotic stress or severe borer damage cutting off the stalk; slow, spreading patches point to soil-borne pathogens or nematodes).
- **Are the symptoms distributed uniformly across the entire field, or are they appearing in isolated patches or low-lying areas?** (Uniform yellowing suggests nitrogen deficiency or widespread drought; isolated, stunted patches point to white grubs, nematodes, or localized wilt pockets).
- **Are the affected plants close to water channels or in waterlogged depressions?** (Prolonged standing water suffocates roots and creates microclimates highly conducive to fungal infections like Wilt and Rust).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- **Look closely at the leaves: Are there long, narrow yellow/white stripes, or orange-brown pustules on the underside?** (Distinct pencil-line stripes indicate Leaf Scald or Mosaic Virus; raised rust-colored spots confirm Sugarcane Rust; distorted, shortened, and crinkled top leaves point to Pokkah Boeng).
- **Look at the growing point (top of the cane): Is there a black, dusty, whip-like structure emerging from the center?** (This is the absolute classic diagnostic indicator for Sugarcane Smut).
- **If you cut a drying or yellowing stalk and split it vertically, what does the internal tissue look and smell like?** (If the internal tissue is reddened with distinct *transverse white bands/patches* and smells sour or alcoholic, this confirms Red Rot. If it is hollow, muddy-colored, and stringy without the sour smell, it indicates Sugarcane Wilt. Clean, white tissue suggests borer damage or drought).
- **Where are the symptoms starting?** (Drying of the central whorl/topmost leaves often points to Top Borer damage or Pokkah Boeng; premature drying of lower leaves can be normal senescence or a sign of severe Wilt).

Category IV: Water, Soil & Root Architecture

- **What is the irrigation frequency, and has the field experienced waterlogging for extended periods?** (Sugarcane requires high water but zero stagnation. Standing water induces root anoxia, mimics yellowing diseases, and severely exacerbates Wilt).
- **What is the typical pH and EC (salinity) of the soil?** (Calcareous or high pH soils frequently cause severe Iron/Zinc chlorosis in sugarcane, which manifests as striking white or yellow interveinal striping on new leaves, easily confused with viral infections).
- **If you dig up a stunted plant, what does the root system look like?** (Thick, stubby, or blackened roots lacking fine feeder roots indicate nematode damage or soil-borne root rots; chewing damage at the base indicates termites or white grubs).

Category V: Crop Nutrition & Chemical Input History

- **What is your basal and top-dressing fertilizer history (Urea, DAP, Potash)?** (Excessive late-season nitrogen makes the cane succulent, increasing vulnerability to borers, Pyrilla (leafhoppers), and fungal diseases. Lack of potash reduces the physical stalk strength, leading to lodging and subsequent rot).
- **Have you applied any pre- or post-emergence weedicides recently?** (Improper application of herbicides like Atrazine or 2,4-D can cause temporary leaf scorching, chlorosis, or abnormal tillering that looks like a physiological disease).

Category VI: Environmental Microclimate & Pest Vectors

- **What has the weather been like over the past 3 to 4 weeks?** (High humidity, heavy monsoon rains, and strong winds are the primary drivers for the rapid spread of Red Rot, Rust, and Smut spores).
- **Are there signs of insect damage, specifically borer exit holes or heavy infestations of Pyrilla/Whiteflies?** (Borers create physical wounds in the stalk that act as direct entry points for Red Rot and Wilt fungi. Heavy Pyrilla feeding excretes honeydew, causing black Sooty Mold to cover the leaves, severely reducing photosynthesis).

Category VII: Historical Cultivation & Previous Actions

- **How many times has this specific field been ratooned?** (Fields kept for 3rd or 4th ratoons harbor massive build-ups of soil pathogens, nematodes, and systemic diseases like RSD and Smut).
- **Did you treat the seed-cane (setts) before planting? Specifically, did you use a hot water treatment or a fungicide dip?** (Hot water treatment is the only effective way to eliminate deep-seated systemic pathogens like RSD and Smut; fungicide dips protect against early soil-borne rots. Lack of treatment is a major red flag).
- **Have you applied any curative chemicals or granular insecticides (like Carbofuran/Fipronil) recently?** (Knowing this rules out chemical toxicity and helps determine if the current symptoms are a continuation of an untreated problem or a failure of the applied treatment).

RICE DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In rice pathology, field diagnostics are unique because the crop is traditionally grown under flooded or saturated conditions, which heavily influences microclimate and root health. An accurate diagnosis requires distinguishing biotic pathogens (fungi, bacteria, viruses, and nematodes) from abiotic stresses (soil salinity, alkalinity, zinc deficiency, or herbicide toxicity). Agronomists analyze macro-field distribution patterns down to individual leaf lesions and stem dissections before forming a management strategy.

2. Agronomist Investigative Questionnaire

Category I: Variety, Crop Phenology & Establishment (The Genetic & Structural Baseline)

- **What specific rice variety is sown (e.g., Fine/Basmati vs. Coarse/Hybrid), and what was the planting method?** (Basmati varieties are historically highly susceptible to Leaf/Neck Blast and Bakanae, while certain hybrids are highly vulnerable to Bacterial Leaf Blight. Direct Seeded Rice [DSR] faces different initial soil/weed stresses compared to traditional puddled transplantation).
- **What is the exact age of the crop, or how many Days After Transplanting (DAT) or Days After Sowing (DAS) is it?** (Seedling/nursery stage is prone to Damping-off and Bakanae; maximum tillering to panicle initiation favors Sheath Blight and Blast; heading/flowering stages are critical for Neck Blast and False Smut).
- **What was the condition of the nursery before pulling?** (Yellowish, weak seedlings or exceptionally tall, lanky seedlings in the nursery bed point to early infection or nutritional stress carried over to the main field).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- **How fast are the symptoms developing, and what is the field layout of the damage?** (Rapid yellowing and drying across large, uniform blocks usually point to abiotic issues like sudden salinity flushes or water depth issues. Slow-spreading, circular patches indicate fungal or insect foci).
- **Are there patches where the rice tillers are completely lodged, dry, and brown ("burnt")?** (Circular patches of dried, collapsed plants indicate "hopper burn" caused by heavy Brown Planthopper [BPH] infestations at the plant base, which can be misdiagnosed as Stem Rot or severe drought).
- **Are the symptoms localized along the field borders or water inlets?** (Border-heavy yellowing or stunting often points to viral vectors moving in from wild grasses on dikes; damage concentrated near water inlets can indicate chemical toxicity or cold water stress).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- **Look closely at the leaf blades: What is the shape, color, and margin of the lesions?**
 - *Spindle/diamond-shaped spots* with grey centers and brown borders indicate **Rice Blast**.
 - *Wavy, water-soaked streaks* starting from the leaf tip or margins and turning paper-white indicate **Bacterial Leaf Blight (BLB)**.
 - *Small, regular, oval brown spots* evenly distributed across leaves indicate **Brown Spot** (often an indicator of poor soil nutritional status).

- **Examine the leaf sheaths just above the water line: Do you see greenish-grey, oblong lesions with irregular borders?** (Large, "snake-skin" or banded patterns on the lower sheaths confirm **Sheath Blight**).
- **Examine the stature of individual abnormal plants: Are they twice as tall as neighboring tillers?** (Abnormally elongated, pale, thin tillers that eventually tilt and die confirm **Bakanae / Foolish Seedling disease**).
- **Pull a drying tiller and check the stem: Is the panicle empty and white ("Whitehead"), or is the young central leaf dry and dead ("Deadheart")?** (Split the stem vertically. If it is hollow with a chewing insect or frass inside, it is **Stem Borer** damage. If the stem base is dark, slimy, and has a foul smell without insect larvae, it points to **Bacterial Stem Rot**).
- **Look at the panicle neck during heading: Is there a dark brown/black lesion right at the base of the panicle?** (A snapped or rotted panicle base confirms **Neck Blast**).

Category IV: Water, Soil & Root Architecture

- **What has the water management regime been like (Continuous flooding vs. Alternate Wetting and Drying [AWD])?** (Continuous deep flooding creates high humidity in the lower canopy, rapidly accelerating Sheath Blight. Conversely, allowing the field to completely dry out and experience drought stress sharply triggers **Rice Blast** surges).
- **What is the pH and electrical conductivity (EC) of the soil and irrigation water?** (Rice is sensitive to salinity at the seedling and panicle initiation stages. High salinity causes leaf-tip scorching and poor grain filling).
- **Pull up a whole hill and wash the roots: What color are they and what do they smell like?** (Healthy roots are reddish-brown to white. Jet-black or dark brown roots that smell like rotten eggs indicate severe sulfide toxicity or anoxia due to prolonged, unbroken waterlogging).

Category V: Crop Nutrition & Chemical Input History

- **How many total splits and what exact doses of Nitrogen (Urea) have been applied?** (Heavy, late-stage applications of Nitrogen create a dense, succulent canopy that reduces light penetration and drastically increases the crop's susceptibility to Leaf Blast, Sheath Blight, and BLB).
- **Has Zinc Sulfate been applied, and what color are the older leaves?** (In high pH or continuously flooded soils, Zinc becomes unavailable. This causes **Khaira disease** [Zinc deficiency], characterized by rusty-brown or bronzed streaks on older leaves, appearing two to four weeks after transplanting).
- **What herbicides were applied, and at what crop stage?** (Overdoses or late applications of certain post-emergence herbicides can cause severe yellowing, leaf rolling, or temporary stunting that mimics viral diseases).

Category VI: Environmental Microclimate & Pest Vectors

- **What has the weather been like over the past week (temperature, rainfall, humidity)?** (Cool nights, warm days, and heavy morning dew create the ideal window for **Blast** spores to germinate. High winds, heavy monsoon rains, and splashing water cause mechanical injury to leaves, allowing **BLB** bacteria to enter and spread rapidly).
- **Part the canopy and check the base of the hills: Are there active populations of planthoppers or leafhoppers?** (Brown Planthoppers [BPH] damage the crop directly via feeding and vectoring ragged stunt virus; Green Leafhoppers [GLH] serve

as the primary vector for **Tungro Virus**, which causes severe stunting and bright yellow-orange discoloration of leaves).

Category VII: Historical Cultivation & Previous Actions

- **Was the seed treated before nursery sowing, and how was the field prepared?** (Fungicidal seed soaking significantly drops early-season Blast and Bakanae pressure. Incorporating or burning infected stubble from the previous season reduces the primary inoculum for Stem Borers and Stem Rot).
- **Have any preventative or curative fungicides (like Tricyclazole, Azoxystrobin, or copper-based compounds) already been sprayed?** (Assesses if the pathogen has developed localized tolerance, evaluates chemical timing, and prevents phytotoxicity from unnecessary tank-mixes).

MAIZE (CORN) DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In maize pathology, an accurate field-level diagnosis requires isolating biotic pathogens (fungi, bacteria, viruses, and nematodes) from abiotic conditions (zinc or nitrogen deficiencies, drought stress, or herbicide carryover). Maize is highly sensitive to early-stage moisture stress and canopy microclimates. Agronomists map macro field layouts down to leaf lesion morphology, stalk integrity, and ear development before formulating targeted remediation strategies.

2. Agronomist Investigative Questionnaire

Category I: Variety & Crop Phenology (The Genetic & Structural Baseline)

- **What specific maize hybrid or variety was planted, and is it a spring or autumn crop?** (Spring and autumn crops face vastly different temperature and pest dynamics; certain hybrids have high built-in resistance to Turcicum Blight but may be vulnerable to stalk rots).
- **What is the exact age of the crop, or what is its current vegetative/reproductive growth stage ?** (Seedling stages are vulnerable to Pythium damping-off; rapid vegetative growth triggers Downy Mildew or Leaf Blights; post-flowering/grain-fill stages are critical windows for Stalk Rots and Ear Rots).
- **What was the plant population density achieved per acre?** (Over-crowded plant populations alter the canopy microclimate, trapping humidity and exponentially increasing the severity of leaf blights and stalk lodging).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- **How quickly did the symptoms appear, and what is their physical footprint across the field?** (A sudden, uniform bleaching or burning across the entire field indicates chemical drift or extreme weather. Scattered, expanding patches indicate biological pathogens like fungal rots or nematode complexes).
- **Are the symptoms concentrated in low-lying spots with poor drainage or heavy clay content?** (Poorly drained pockets favor water-loving soil fungi like *Pythium* and cause temporary root anoxia, mimicking nitrogen starvation).
- **Are the symptomatic plants tracking along specific rows?** (Row-specific damage heavily indicates mechanical issues, localized tractor compaction, or uneven fertilizer/herbicide application rather than an infectious disease).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- **Look closely at the leaves: What is the precise geometry and color of the lesions?**
 - *Large, long, cigar-shaped (spindle-shaped) grayish-green or tan lesions* indicate **Turcicum Leaf Blight (Northern Corn Leaf Blight - NCLB)**.
 - *Small, narrow, rectangular lesions* restricted between parallel leaf veins indicate **Maydis Leaf Blight (Southern Corn Leaf Blight - SCLB)** or **Gray Leaf Spot (GLS)**.
 - *Widespread chlorotic stripping or downy white growth* on the leaf undersides points to **Downy Mildew**.

- **Examine the stalks of lodging or wilting plants: Push or pinch the lower internodes.**
 - If the lower stalk is easily crushed or collapses when pushed at a 45-degree angle, perform a vertical split.
 - *Disintegrated, shredded internal pith turned pinkish-red* confirms **Gibberella or Fusarium Stalk Rot**.
 - *Internal pith showing a dark, ash-gray discoloration with tiny black speckles (sclerotia)* confirms **Charcoal Rot**.
- **Examine the ears, silks, and tassels: Do you see abnormal growths or molds?**
 - *Massive, distorted galls/swellings* covered in a silvery membrane that ruptures to release black, powdery spores confirm **Corn Smut (*Ustilago maydis*)**.
 - *Whitish-pink mold growing over or between the kernels* points to **Fusarium Ear Rot**, whereas a *greenish-blue mold* suggests **Penicillium or Aspergillus**.

Category IV: Water, Soil & Root Architecture

- **What is the current irrigation schedule, and did the field experience standing water at the early vegetative stage?** (Maize cannot tolerate standing water; even 24-48 hours of saturation at the V2-V5 stage stunts the root network permanently, causing purple leaf margins due to restricted phosphorus uptake).
- **What is the pH and electrical conductivity (EC) of the soil?** (Maize performance drops in highly saline-sodic soils. High pH environments bind up micronutrients, causing prominent physiological chlorosis).
- **Dig up the root mass and slice the primary roots: What do you see?** (Healthy roots are firm and creamy white. Waterlogged or infected roots turn mushy and water-soaked, while chewed or pruned root tips reveal **Rootworm** or **White Grub** activity).

Category V: Crop Nutrition & Chemical Input History

- **What are the exact splitting doses used for Nitrogen (Urea)?** (Maize is a heavy feeder. Nitrogen deficiency shows as a classic **V-shaped yellowing** starting at the leaf tip and moving down the midrib on older leaves. Conversely, excess nitrogen creates succulent stalks prone to lodging and bacterial stalk rots).
- **Was Zinc applied during land preparation or as a foliar spray?** (Zinc deficiency is incredibly common in high pH/calcareous soils, showing up as "**White Bud**"—broad, bleached white-to-yellow bands on either side of the midrib on young leaves).
- **What post-emergence herbicides were used (e.g., Atrazine, 2,4-D, or Glyphosate)?** (Misapplied or late applications of 2,4-D cause brittle stalks, leaf "onion-skinning" or rolling, and fused prop roots easily mistaken for a viral or genetic deformity).

Category VI: Environmental Microclimate & Pest Vectors

- **What has the microclimate been like within the canopy over the past two weeks?** (Moderate temperatures [20-25°C] combined with prolonged morning dew or high humidity drive Turcicum Blight surges. Hot, dry winds favor Charcoal Rot progression).
- **Are there active insect populations, specifically stem borers, armyworms, or aphids?** (Maize Stem Borer larvae drill directly into stalks, creating exit holes that serve as the main gateway for fungal stalk rots. Heavy aphid infestations vector

Maize Dwarf Mosaic Virus (MDMV) and lead to extensive Sooty Mold on leaves and tassels).

Category VII: Historical Cultivation & Previous Actions

- **Was maize planted directly after maize (monoculture), or was it rotated?**
(Continuous maize cultivation causes a compounding build-up of overwintering fungal inocula like Gray Leaf Spot and Turcicum in the surface crop residue).
- **What tillage method was used?** (Minimum or zero-tillage systems retain more surface crop residue, which acts as a primary launchpad for airborne fungal spores if the previous crop was diseased).
- **Have any preventative triazole or strobilurin-based fungicides been applied?**
(Establishes whether the current problem is a case of fungicide mistiming, incorrect dosage, or a breakthrough infection requiring an alternate chemical class).

WHEAT DISEASE DIAGNOSTIC PROTOCOL

1. The Diagnostic Framework

In wheat pathology, accurate field-level diagnosis requires separating biotic pathogens (fungi, bacteria, viruses, and nematodes) from critical abiotic constraints (such as terminal heat stress, soil salinity, frost damage, or nitrogen starvation). Because wheat is a high-density, field-scale crop often grown in distinct thermal windows, agronomists analyze macro-field infestation lines down to spikelet anatomy and pustule structural geometry before deploying a control strategy.

2. Agronomist Investigative Questionnaire

Category I: Variety & Crop Phenology (The Genetic & Structural Baseline)

- **What specific wheat variety was sown, and what is its historical resistance profile?** (Crucial for predicting resistance breakdown to localized virulence shifts, particularly with Stripe and Leaf Rust races).
- **What is the exact growth stage of the crop (e.g., Tillering, Jointing, Booting, Heading, Milking, or Dough stage)?** (Early tillering is vulnerable to seedling blights; jointing to booting is the primary window for Yellow Rust expansion; heading and grain-fill are critical for Leaf Rust, Loose Smut, and localized Head Blight).
- **What was the sowing date?** (Late-sown wheat shifts the grain-filling window directly into high-temperature spring months, inducing terminal heat stress and altering the late-season pest/pathogen pressure).

Category II: Spatial & Chronological Patterns (Macro-Field Level)

- **What is the physical footprint of the symptoms across the field?** (Rusts typically appear first in localized "hotspots" or focal patches oriented with prevailing wind directions before exploding across the canopy. Widespread, perfectly uniform bleaching across the field tips suggests frost injury or severe salinity flushes).
- **Are the symptoms tracking along low-lying, damp, or shaded zones of the field?** (High canopy humidity in low spots or field sections shaded by trees heavily accelerates Powdery Mildew and Stripe Rust development).
- **Is there a pattern of patchy, stunted growth where plants are drying prematurely?** (Circular or irregular patches of premature drying often point to soil-borne root or crown rots rather than foliar disease).

Category III: Symptom Anatomy & Tissue Analysis (Micro-Plant Level)

- **Look closely at the leaves: What is the pattern, geometry, and color of the pustules or lesions?**
 - *Bright yellow/orange pustules arranged linearly in distinct parallel rows/stripes along the veins* confirm **Stripe Rust (Yellow Rust)**.
 - *Scattered, round-to-oval orange-brown pustules* distributed randomly across the leaf surface without a row pattern indicate **Leaf Rust (Brown Rust)**.
 - *Elongated, dark-red to black pustules* breaking through the epidermis of both the leaf sheaths and the stems indicate **Stem Rust (Black Rust)**.
- **Examine the emerging spikes (heads): Is the grain architecture normal?**

- If the entire spikelet structure is completely replaced by a charcoal-black, loose, powdery mass of spores that blows away in the wind, leaving a bare central axis (rachis), confirm **Loose Smut**.
- If the head appears normal but individual grains are swollen, dull-green, and leak a black, greasy paste with a foul, fishy odor when crushed, confirm **Karnal Bunt**.
- **Examine the lower stems and crown: Are there dark lesions or discoloration?** (Brownish, dry rot at the base of the stem and crown indicates **Fusarium Crown Rot** or Bipolaris root rot, distinct from physiological drying).

Category IV: Water, Soil & Root Architecture

- **What is the irrigation regime, and did water log the field during the jointing or booting stages?** (Saturated soils suffocate root tips, leading to a temporary yellowing of the lower leaves that mimics nitrogen or sulfur deficiencies).
- **What is the soil pH and salinity (EC) profile?** (High pH or saline patches cause leaf tip scorching, reduce spikelet fertility, and limit nutrient absorption, which can mask or mimic early root rot stresses).
- **Dig up a sample plant and wash the root system thoroughly: What do the roots look like?** (Healthy roots are extensive and creamy white. If roots are short, blackened, brittle, or show a dark "shiny" rot extending into the lower stem, it points to a complex of **Take-all** or **Fusarium Root Rot**).

Category V: Crop Nutrition & Chemical Input History

- **What is the total quantity and timing of Nitrogen applications?** (Excessive or delayed nitrogen inputs create a lush, soft, succulent canopy that reduces light penetration and dramatically favors the rapid spread of rust fungi and Powdery Mildew).
- **Has there been a balanced application of Phosphorus and Potassium?** (Phosphorus deficiency causes purpling of the older leaf blades and weak tillering; potassium deficiency compromises cell wall structural integrity, increasing vulnerability to lodging and stem pathogens).
- **Were any hormonal herbicides (like 2,4-D) sprayed late?** (Applying 2,4-D past the jointing stage causes severe spikes and head deformities, including malformed or fused spikelets, which can mimic viral or genetic abnormalities).

Category VI: Environmental Microclimate & Pest Vectors

- **What has the weather been like over the past 10 days?** (Cool temperatures [10–20°C] paired with continuous morning fogs or intermittent rains are the ultimate catalysts for **Stripe Rust**. A sudden shift to warm, humid weather [15–28°C] suppresses Stripe Rust but triggers **Leaf Rust** surges).
- **Are there active colonies of aphids or leafhoppers within the crop?** (High aphid pressure causes direct leaf yellowing via sap feeding and leaves honeydew that leads to Sooty Mold. Certain species also vector **Barley Yellow Dwarf Virus (BYDV)**, which causes distinct, bright yellow or purple leaf tipping on individual tillers).

Category VII: Historical Cultivation & Previous Actions

- **What was the previous crop grown in this field during the summer (Kharif) season?** (Wheat sown directly after paddy rice can experience higher **Fusarium** inoculum pressure, as the fungus can survive on rice stubble, contributing to crown or head blights under the right conditions).

- **Was the seed treated with a systemic fungicide prior to sowing?** (Systemic seed treatments are highly effective at completely suppressing **Loose Smut** and protecting seedlings from early soil-borne rots. A lack of seed treatment is a primary reason for smut breakthroughs).
- **Have any preventative triazole or strobilurin fungicides already been applied?** (Assesses whether a current infection represents a failure of chemical timing, poor coverage, or an infection that occurred after the residual protection window of a previous spray had expired).